Fast-QEP: end-to-end result audit

Cell segmentation | Experiment ID 20261015 | Reviewed 14 Oct 2026, America/Phoenix

Finding: the prescribed q=1.5 posterior-mean substitution produced small reconstruction and foreground changes, but no AP50/AP75 or merge/split improvement on either development image. This is an application negative for this fixed configuration, not a refutation of QEP theory.

Primary comparison: original R downstream held fixed

Image / arm	AP50	AP75	Merge / split	FG Dice
Nuclei / Fast-GP	0.64810127	0.28402367	22 / 1	0.85039341
Nuclei / Fast-QEP	0.64810127	0.28402367	22 / 1	0.85029522
Whole-cell / Fast-GP	0.58386412	0.17111460	68 / 12	0.85841116
Whole-cell / Fast-QEP	0.58386412	0.17111460	68 / 12	0.85839488

AP here is the frozen legacy TP/(TP+FP+FN) score, with per-GT-best-IoU matching. It is not precision-recall curve area. Primary QEP is the pooled four-chain estimate; individual chains are diagnostics.

Why equal AP does not mean equal images

Quantity	Nuclei	Whole-cell
Mean absolute reconstruction difference (0-255 units; estimated QEP mean)	0.0053725	0.0022274
Foreground pixels changed vs Fast-GP	121 / 1,085,136 (0.01115%)	7 / 361,200 (0.001938%)
Gained / lost matched GT at IoU 0.5	0 / 0	0 / 0
TP / FP / FN at IoU 0.5 (same in both primary arms)	256 / 65 / 74	275 / 68 / 128

The mean differences are area-weighted from all tile summaries, including the unchanged two trailing columns. They include finite Monte Carlo error. Initial/final threshold proportions and outlier flags are unchanged in all 25 pooled-QEP tiles; intensity thresholds change slightly because tile maxima change. The new masks are not identical to Fast-GP masks, but no GT crosses the AP50 matching decision. Nuclei spurious count falls 23 to 22 while foreground Dice falls slightly; this is not a merge-reduction gain.

Inference and baseline checks

Recorded diagnostic	Nuclei (16 tiles)	Whole-cell (9 tiles)
Worst monitored Rhat	1.000925	1.000872
Minimum bulk / tail ESS	8089.9 / 7399.2	7812.5 / 5628.5
Largest pixel MCSE (0-255)	0.000659	0.002177
Sampling time only	88.62 min	20.95 min

All 25 recorded gates and prescribed seed/geometry checks pass; no recorded retained divergence, nonfinite HMC proposal or global numerical failure. Total sampling time: 109.58 min. Actual R baseline records match historical masks/partitions; Python q=2 vs R maximum normalized error is 8.25e-15 / 1.13e-13. These are monitored diagnostics, not a mathematical proof of mixing.

The exceptional chain and the decision

A threshold-selection sensitivity finding, not a selected positive QEP result

Nuclei chain 3 is the only chain with an AP gain

Quantity	Fast-GP / pooled QEP	QEP chain 3
AP50	0.64810127	0.65561224
AP75	0.28402367	0.30060120
Merge / split	22 / 1	23 / 1
Matched GT changes vs Fast-GP	Pooled: gain 0, lose 0	Gain 4, lose 3
Changed foreground vs Fast-GP	Pooled: 121 pixels	5,715 pixels
Tile 9 initial / final proportion	0.76 / 0.76	0.69 / 0.72
Tile 9 components before outlier repair (R count includes background)	30	199
Outlier tiles	Tile 5 only	Tiles 5 and 9

The added outlier also changes tile 5's final proportion from 0.7226667 to 0.72. All outlier flags, replacement-proportion arithmetic and p x maximum intensity thresholds were checked across the 16 supplied threshold tables. Chain 3 gains GT labels 7, 9, 10 and 67, but loses 54, 61 and 86. Its maximum threshold-intensity shift vs Fast-GP is about 9.98 units, versus 0.0528 for pooled QEP.

Interpretation: these records localize the exceptional result to threshold selection and outlier handling. They support sensitivity of the downstream decisions to chain-mean perturbations. The criterion landscape and masks were not supplied, so a controlled causal attribution is not established. Low pixel MCSE alone does not guarantee stability of discrete thresholding or AP. Three other nuclei chains tie the baseline; every whole-cell chain ties it. Choosing chain 3 would violate the frozen primary estimator.

What the Gaussian control establishes

The fixed g_eta Gaussian smoothing-strength control ties QEP on AP50/AP75 and merge/split in both images; its nuclei endpoint metrics also coincide. This provides no QEP-specific advantage at these endpoints. It does not establish exact equality between GP and QEP posterior distributions or reconstruction arrays.

Decision and limits

Freeze this branch as a negative result: fixed R hyperparameters + covariance-matched finite-tile qED prior + independent Gaussian noise + q=1.5 posterior mean + unchanged original R downstream did not improve the prescribed instance endpoints. Repeating this configuration or selecting q/seeds after observing AP has no current justification. If Fast-QEP work continues, first justify a structural mechanism that can add useful spatial separation evidence beyond a Gaussian smoothing-strength change. This audit starts no new experiment.

Scope: one development image per dataset; no significance, equivalence-test or held-out generalization claim. The lightweight archive omits arrays, chains, source files and detailed R logs. This is a consistency audit of supplied records, not independent recomputation of masks, IoU or convergence statistics from raw draws. All 72 uploaded files covered by the completion manifest match their recorded hashes; 214 manifest-listed files were not supplied. Sixteen AP50/AP75 rows were recomputed from counts; runtime signatures and all 25 checkpoint summaries agree.

Sources: metrics.csv; comparisons.json; MC_stability.json; environment.json; completion.json; run.log; preflight/R_anchor_checks.json; preflight/preflight_complete.json; all uploaded tile summaries/checkpoints and threshold tables. Numeric audit details are attached to this PDF as audit.json.